ADVANCED LINEAR ALGEBRA
HOMEWORK 3

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P.1.32 (a) Give an example of a nonzero $A \in \mathbf{M}_2(\mathbb{F})$ such that $\text{col}(A) = \text{null}(A)$. (b) Give an example of $A \in \mathbf{M}_2(\mathbb{F})$ such that $\text{col}(A) \oplus \text{null}(A) = \mathbb{F}^2$.

Solution. Sol □

P.1.39 In $\mathbf{M}_n$, let $\mathcal{U}$ denote the set of strictly lower triangular matrices and let $\mathcal{W}$ denote the set of strictly upper triangular matrices. (a) Show that $\mathcal{U}$ and $\mathcal{W}$ are subspaces of $\mathbf{M_n}$. (b) What is $\mathcal{U} + \mathcal{W}$? Is this sum a direct sum? Why?

Solution. Sol □

P.1.41 Let $f \in C[-1,1]$. Then if f is *even* if $f(-x) = f(x)$ for all $x \in [-1,1]$, and f is *odd* if $f(-x) = -f(x)$ for all $x \in [-1,1]$. Let $\mathcal{U}$ denote the set of even functions in $C[-1,1]$ and let $\mathcal{V}$ denote the set of odd functions in $C[-1,1]$. Show that (a) $\mathcal{U}$ and $\mathcal{V}$ are subspaces of $C[-1,1]$, and (b) $C[-1,1] = \mathcal{U} \oplus \mathcal{W}$. *Hint.* $f(x) = \frac{1}{2}(f(x) + f(-x)) + \frac{1}{2}(f(x) - f(-x))$.

Solution. Sol □

P.1.44

Solution. Sol □

P.2.1

Solution. Sol □

P.2.2

Solution. Sol □

P.2.3

Solution. Sol □

P.2.4

Solution. Sol □

P.2.5

Solution. Sol □

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P.2.6

Solution. Sol ∎